1. (a) Consider the real-coefficient transfer function

$$H_M(z) = \frac{\sum_{n=0}^{n=M} b[n]z^{-n}}{\sum_{n=0}^{n=M} a[n]z^{-n}}, M > 0$$

(i) Prove that when the coefficients of the numerator and denominator satisfy the symmetry condition $a[n] = b[M-n]$, the transfer function $H_M(z)$ corresponds to an all-pass filter. **[3]**

Answer

The condition $|H_M(z)| = 1$, for $|z| = 1$ implies $\left|H_M\left(e^{j\omega}\right)\right| = 1$ for all ω. That means we are looking for an allpass transfer function. Consider an Mth order causal, real-coefficient all pass transfer function of the form

$$H_M(z) = \frac{\sum_{n=0}^{n=M} a[M-n]z^{-n}}{\sum_{n=0}^{n=M} a[n]z^{-n}}$$

If we denote the denominator polynomial of the allpass function as $A_M(z)$ with

$$A_M(z) = \sum_{n=0}^{n=M} a[n]z^{-n}$$

then it follows that $H_M(z)$ can be written as:

$$H_M(z) = \frac{z^{-M}A_M(z^{-1})}{A_M(z)}$$

From the above we see that:

$$H_M(z^{-1}) = \frac{z^{M}A_M(z)}{A_M(z^{-1})}$$

$$H_M(z)H_M(z^{-1}) = \frac{z^{-M}A_M(z^{-1})}{A_M(z)}\frac{z^{M}A_M(z)}{A_M(z^{-1})} = 1 \Rightarrow \left|H_M\left(e^{j\omega}\right)\right|^2 = H_M(z)H_M(z^{-1}) = 1$$

$$\Rightarrow \left|H_M\left(e^{j\omega}\right)\right| = 1.$$

Therefore, the condition $b[n] = a[M-n]$ corresponds to an all-pass filter.

(ii) Comment on the locations of the zeros and the poles of a causal, real, stable, allpass filter's transfer function within the $z-$plane, with respect to the unit circle. Justify your answer. **[3]**

Answer

The poles of a causal stable transfer function must lie inside the unit circle. As a result, all zeros of a causal stable allpass transfer function lie outside the unit circle in a mirror-image symmetry with its poles situated inside the unit circle.

(iii) Find the part of the phase response of an allpass transfer function due to its zeros, in terms of the part of the phase response due to its poles. **[3]**

Answer

The phase response of transfer function is $\angle H_M\left(e^{j\omega}\right) = \angle e^{-j\omega M} + \angle A_M\left(e^{-j\omega}\right) - \angle A_M\left(e^{j\omega}\right)$

Since the polynomial is real-coefficient we can say that $A_M(e^{-j\omega}) = A_M^*(e^{j\omega})$. We know that $\angle A_M^*(e^{j\omega}) = -\angle A_M(e^{j\omega})$. Therefore, $\angle H_M(e^{j\omega}) = -\omega M - 2\angle A_M(e^{j\omega})$. The phase response of the zeros is $-\omega M - \angle A_M(e^{j\omega})$ therefore is equal to the phase response of the poles minus ωM.

(iv) Explain how multiplying $H_M(z)$ by z^{-N} affects its amplitude and phase responses, and discuss what happens when N becomes large. **[3]**

Answer

The amplitude response is not affected. A linear phase response component is introduced in the phase response. If N becomes large the phase is linearised.

(b) Consider a second-order, causal, real, stable, allpass transfer function $H(z)$.

(i) Provide the expression for $H(z)$. **[3]**

Answer

$H(z) = \frac{q+pz^{-1}+z^{-2}}{1+pz^{-1}+qz^{-2}}$

(ii) Find the phase response associated with $H(z)$. **[3]**

Answer

$H(z) = \frac{q+pz^{-1}+z^{-2}}{1+pz^{-1}+qz^{-2}}$

$$H(z) = \frac{z^{-1}(qz + p + z^{-1})}{z^{-1}(z + p + qz^{-1})} = \frac{(qz + p + z^{-1})}{(z + p + qz^{-1})}$$

$$\angle H(e^{j\omega}) = \angle(qe^{j\omega} + p + e^{-j\omega}) - \angle(e^{j\omega} + p + qe^{-j\omega})$$
$$= \arctan\left(\frac{q\sin(\omega) - \sin(\omega)}{p + q\cos(\omega) + \cos(\omega)}\right) - \arctan\left(\frac{-q\sin(\omega) + \sin(\omega)}{p + q\cos(\omega) + \cos(\omega)}\right)$$
$$= \arctan\left(\frac{q\sin(\omega) - \sin(\omega)}{p + q\cos(\omega) + \cos(\omega)}\right) + \arctan\left(\frac{q\sin(\omega) - \sin(\omega)}{p + q\cos(\omega) + \cos(\omega)}\right)$$
$$= 2\arctan\left(\frac{q\sin(\omega) - \sin(\omega)}{p + q\cos(\omega) + \cos(\omega)}\right)$$

(c) (i) Using a first-order non-minimum-phase IIR filter, demonstrate that it can be expressed as the product of a minimum-phase IIR filter and an all-pass filter. **[4]**

Answer

Consider a first-order non–minimum-phase IIR filter with a zero outside the unit circle and a pole inside the unit circle:
$H(z) = (1 - qz^{-1})/(1 - pz^{-1})$ where $|q| > 1$ and $|p| < 1$.

The zero at $z = q$ makes $H(z)$ non–minimum-phase, while the pole ensures stability since $|p| < 1$.
Show that $H(z)$ can be expressed as the product of a minimum-phase IIR filter $H_{\min}(z)$ and an all-pass filter $A(z)$.
Reflect the zero outside the unit circle ($|q| > 1$) to inside by defining $a = \frac{1}{q}$. Then define:
$H_{\min}(z) = (1 - (\frac{1}{q})z^{-1})/(1 - pz^{-1})$
This $H_{\min}(z)$ is minimum-phase since its zero lies inside the unit circle $|a| < 1$.
Now form the residual factor:

$$A(z) = \frac{H(z)}{H_{\min}(z)} = \frac{1 - qz^{-1}}{1 - \left(\frac{1}{q}\right)z^{-1}} = q\,\frac{1 - qz^{-1}}{q - z^{-1}}$$

We see that $A(z)$ is an allpass filter.

Generalisation
For higher-order or complex-valued systems, each zero outside the unit circle can be paired with an all-pass term. Replacing all outside zeros by their reciprocal reflections yields a minimum-phase filter, and the product of the all-pass factors accounts for the phase difference.

(ii) Compare the average group delay of a first-order non-minimum-phase IIR filter with that of its minimum-phase counterpart. **[3]**

Answer

Average group delay of a first-order non-minimum-phase IIR filter: $1 - 0 = 1$
Average group delay of a first-order minimum-phase IIR filter: $1 - 1 = 0$

2. (a) Show that a Type III linear-phase, real, FIR transfer function $H(z)$ of length $2M+1$ can be expressed as follows:

$$H(z) = z^{-M}\left(\sum_{k=0}^{M-1} h[k](z^{M-k} - z^{k-M})\right)$$

[4]

Answer

$$H(z) = \sum_{n=0}^{2M} h[n]z^{-n} = \sum_{n=0}^{M-1} h[n]z^{-n} - \sum_{n=M+1}^{2M} h[n]z^{-n}$$

$$= \sum_{n=0}^{M-1} h[n]z^{-n} - \sum_{k=0}^{M-1} h[2M-k]z^{k-2M}$$

$$= \sum_{n=0}^{M-1} h[n]z^{-n} - \sum_{k=0}^{M-1} h[k]z^{k-2M}$$

$$= \sum_{k=0}^{M-1} h[k]z^{-k} - \sum_{k=0}^{M-1} h[k]z^{k-2M}$$

$$z^{-M}\left(\sum_{k=0}^{M-1} h[k](z^{M-k} - z^{k-M})\right)$$

We have proved the required relationship.

(ii) Determine the phase of the expression enclosed in the brackets of equation (1) over the interval $0 < \omega < \pi$. **[4]**

Answer

$$\sum_{k=0}^{M-1} h[k]\left(z^{M-k} - z^{k-M}\right) = \sum_{k=0}^{M-1} h[k]2j\sin(\omega(M-k))$$

Answer is $\frac{\pi}{2}$.

(b) By using the relation

$$z^r + z^{-r} = 2T_r\left(\frac{z+z^{-1}}{2}\right)$$

where $T_r(x)$ is the r −th order Chebyshev polynomial in x, express $H(z)$ of (a) above in the form

$$H(z) = z^{-M}\sum_{n=0}^{M} a[n]\left(\frac{z+z^{-1}}{2}\right)^n \quad (1)$$

in the case where $M = 4$. Determine the relation between $a[n]$ and $h[n]$ for $M = 4$.

[8]

Answer

For $M = 4$ we have

$$H(z) = z^{-4}\left(\sum_{k=0}^{3} h[k]\left(z^{4-k} - z^{k-4}\right)\right)$$

$= z^{-4}\big(h[3](z - z^{-1}) + h[2](z^2 - z^{-2}) + h[1](z^3 - z^{-3}) + h[0](z^4 - z^{-4})\big)$

$T_r(x) = 2xT_{r-1}(x) - T_{r-2}(x), r \geq 2$

$T_0(x) = 1, T_1(x) = x, T_2(x) = 2x^2 - 1$

$T_3(x) = 2xT_2(x) - T_1(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x$

$T_4(x) = 2xT_2(x) - T_1(x) = 2x(2x^2 - 1) - x = 8x^4 - 8x^2 + 1$

$$H(z) = z^{-4}\left(h[3](z - z^{-1})U_0\left(\frac{z + z^{-1}}{2}\right) + h[2](z - z^{-1})U_1\left(\frac{z + z^{-1}}{2}\right) + h[1](z - z^{-1})U_2\left(\frac{z + z^{-1}}{2}\right) + h[0](z - z^{-1})U_3\left(\frac{z + z^{-1}}{2}\right)\right)$$

$$H(z) = z^{-4}(z - z^{-1})\left(h[3]U_0\left(\frac{z + z^{-1}}{2}\right) + h[2]U_1\left(\frac{z + z^{-1}}{2}\right) + h[1]U_2\left(\frac{z + z^{-1}}{2}\right) + h[0]U_3\left(\frac{z + z^{-1}}{2}\right)\right)$$

$$= z^{-4}(z - z^{-1})\left(h[3] + 2h[2]\left(\frac{z + z^{-1}}{2}\right) + h[1]\left(4\left(\frac{z + z^{-1}}{2}\right)^2 - 1\right) + h[0]\left(8\left(\frac{z + z^{-1}}{2}\right)^3 - 4\left(\frac{z + z^{-1}}{2}\right)\right)\right)$$

$$= z^{-4}(z - z^{-1})\left((\boldsymbol{h[3] - h[1]}) + (\boldsymbol{2h[2] - 4h[0]})\left(\frac{z + z^{-1}}{2}\right) + \boldsymbol{4h[1]}\left(\frac{z + z^{-1}}{2}\right)^2 + \boldsymbol{8h[0]}\left(\frac{z + z^{-1}}{2}\right)^3\right)$$

The Chebyshev polynomials of second kind satisfy the following recursive relationship:

$U_r(x) = 2xU_{r-1}(x) - U_{r-2}(x), r \geq 2$

$U_0(x) = 1, U_1(x) = 2x$

(c) (i) Show that a linear-phase FIR filter with symmetric impulse response and length M satisfies the mirror image polynomial relationship given below. **[3]**

$$H(z) = z^{-(M-1)}H(z^{-1})$$

Answer

Let the FIR filter have an impulse response $h[n]$ of length M.

$$H(z) = \sum_{n=0}^{M-1} h[n]z^{-n}$$

$$z^{-(M-1)}H(z^{-1}) = z^{-(M-1)}\sum_{n=0}^{M-1} h[n]z^{n}$$

$$= z^{-(M-1)}\sum_{r=0}^{M-1} h[M - 1 - r]z^{M-1-r} = \sum_{r=0}^{M-1} h[r]z^{-r} = H(z)$$

(ii) Show that a linear-phase FIR filter with antisymmetric impulse response and length M satisfies the antimirror image polynomial relationship given below. **[3]**

$$H(z) = -z^{-(M-1)}H(z^{-1})$$

Answer

Let the FIR filter have an impulse response $h[n]$ of length M.

$$H(z) = \sum_{n=0}^{M-1} h[n]z^{-n}$$

$$z^{-(M-1)}H(z^{-1}) = z^{-(M-1)} \sum_{n=0}^{M-1} h[n]z^{n}$$

$$= z^{-(M-1)} \sum_{r=0}^{M-1} h[M-1-r]z^{M-1-r} = -\sum_{r=0}^{M-1} h[r]z^{-r} = -H(z)$$

(iii) Using the Mirror-Image Polynomial (MIP) and Anti–Mirror-Image Polynomial (AIP) relationships, show that band-stop filters can only be realised with Type I linear-phase FIR filters. **[3]**

Answer

Type I: $M-1$ is even.
There isn't any relation which indicates that there is zero at $z = 0$ or $z = \pi$.
All types of filters can be designed.

Type II: $M-1$ is odd.
$H(z) = z^{-(M-1)}H(z^{-1})$
$H(-1) = -H(-1) \Rightarrow H(-1) = 0$
A bandstop filter cannot be designed.

Type III: $M-1$ is even.
$H(z) = -z^{-(M-1)}H(z^{-1})$
$H(1) = -H(1) \Rightarrow H(1) = 0$
$H(-1) = -H(-1) \Rightarrow H(-1) =$
A bandstop filter cannot be designed.

Type IV: $M-1$ is odd.
$H(z) = -z^{-(M-1)}H(z^{-1})$
$H(1) = -H(1) \Rightarrow H(1) = 0$
A bandstop filter cannot be designed.

3. (a) The bilinear transformation from the s −plane to the z −plane is given by

$$s = \frac{1 - z^{-1}}{1 + z^{-1}}$$

(i) Explain using mathematical relations, where each point $s = \sigma + j\Omega$ on the s −plane is mapped on the z −plane through the bilinear transformation. **[5]**

Answer

$z = \frac{1+s}{1-s}$

For $s = j\Omega_0$ we have that $z = \frac{1+j\Omega_0}{1-j\Omega_0}$ which has a magnitude equal to 1. This implies that a point on the imaginary axis in the s −plane is mapped onto a point on the unit circle in the z −plane where $|z| = 1$. In the general case, for $s = \sigma_0 + j\Omega_0$

$z = \frac{1+\sigma_0+j\Omega_0}{1-\sigma_0-j\Omega_0} \Rightarrow |z|^2 = \frac{(1+\sigma_0)^2 + {\Omega_0}^2}{(1-\sigma_0)^2 + {\Omega_0}^2}$

A point in the left half s −plane with $\sigma_0 < 0$ is mapped onto a point inside the unit circle in the z −plane as $|z| < 1$. Likewise, a point in the right half s −plane with $\sigma_0 > 0$ is mapped onto a point outside the unit circle in the z −plane as $|z| > 1$.

(ii) Prove that the relationship between the continuous-time angular frequency and the discrete-time angular frequency is non-linear. **[5]**

Answer

We see that

$j\Omega = \frac{1-e^{-j\omega}}{1+e^{-j\omega}} = j\tan\left(\frac{\omega}{2}\right) \Rightarrow \Omega = \tan\left(\frac{\omega}{2}\right)$ which is a non-linear relationship.

(b) A given real lowpass digital IIR filter has a rational transfer function $H_L(z)$ and a cutoff frequency ω_c. The transfer function is transformed by replacing z by $F(z) = \frac{az+b}{cz+d}$, to a real highpass rational transfer function $H_H(z) = H_L(F(z))$ of the same order as $H_L(z)$ but different cutoff frequency. By using the constraints $H_H(e^{j\pi}) = H_L(e^{j0}) = 1$ and $H_H(e^{j0}) = H_L(e^{j\pi}) = 0$, derive relationships among the parameters a, b, c, d and explain what type of filter is $F(z)$. **[5]**

Answer

$H_H(e^{j0}) = H_L\left(F(e^{j0})\right) = H_L(F(1)) = 0$

Therefore, $F(1) = \frac{a+b}{c+d} = e^{j\pi} = -1 \Rightarrow a + b = -c - d$

$H_H(e^{j\pi}) = H_L\left(F(e^{j\pi})\right) = H_L(F(-1)) = 1$

Therefore, $F(-1) = \frac{-a+b}{-c+d} = 1 \Rightarrow -a + b = -c + d$

Therefore, $b = -c$ and $a = -d$

$F(z) = \frac{az+b}{-bz-a} = -\frac{az+b}{bz+a}$

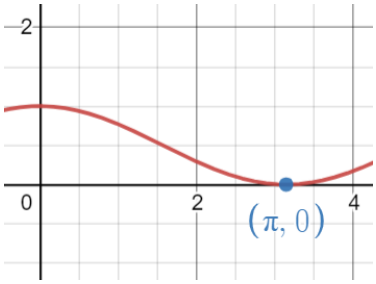
$F(z)$ is an allpass filter.

(c) (i) Consider the Finite Impulse Response (FIR) filter transfer functions $G_1(z) = \frac{1}{4}(z + 2 + z^{-1})$ and $G_2(z) = \frac{1}{4}(-z + 2 - z^{-1})$. Explain what type of filters $G_1(z)$ and $G_2(z)$ represent (e.g. low-pass, high-pass, etc.) and justify your answer. **[5]**

Answer

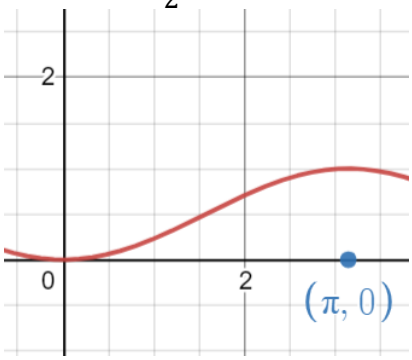
$G_1(z) = \frac{1}{4}(z + 2 + z^{-1}) = \frac{1}{4}(2 + 2\cos(\omega)) = \frac{1}{2}(1 + \cos(\omega))$

$|G_1(e^{j\omega})| = \frac{1}{2}|1 + \cos(\omega)|$. We observe that $|G_1(e^{j\omega})|$ decreases as the frequency approaches π.



$G_2(z) = \frac{1}{4}(-z + 2 - z^{-1}) = \frac{1}{4}(2 - 2\cos(\omega)) = \frac{1}{2}(1 - \cos(\omega))$

$|G_2(e^{j\omega})| = \frac{1}{2}|1 - \cos(\omega)|$



We observe that $|G_2(e^{j\omega})|$ increases as the frequency approaches π.

(ii) Find the average group delay of $G_1(z)$ and $G_2(z)$. **[5]**

Answer

$G_1(z) = \frac{1}{4}(z + 2 + z^{-1}) = \frac{1}{4}z^{-1}(z^2 + 2z + 1) = \frac{1}{4}z^{-1}(z + 1)^2$

Pole at 0. Two zeros at $z = -1$.

$G_1(z) = \frac{1}{4}(-z + 2 - z^{-1}) = -\frac{1}{4}z^{-1}(z^2 - 2z + 1) = -\frac{1}{4}z^{-1}(z - 1)^2$

Pole at 0. Two zeros at $z = 1$.

Average group delay is 0 in both cases. This comes from $1 - \frac{1}{2} - \frac{1}{2} = 0$.

4. (a) Consider the multirate structure of **Figure 1** below, where $H_0(z), H_1(z)$ and $H_2(z)$ are ideal, zero phase, real-coefficient lowpass, bandpass and highpass filters respectively, with frequency responses as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{\pi}{3})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \frac{2\pi}{3})$$

$$H_2(e^{j\omega}) = u(\omega - \frac{2\pi}{3}) - u(\omega - \pi)$$

The function $u(\omega)$ is the well-known unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (-\frac{9\omega^2}{\pi^2} + 1)\left(u(\omega) - u\left(\omega - \frac{\pi}{3}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n]$ and $y_2[n]$.
Notice that all frequency domain representations above are provided only within the range $[0, \pi]$.

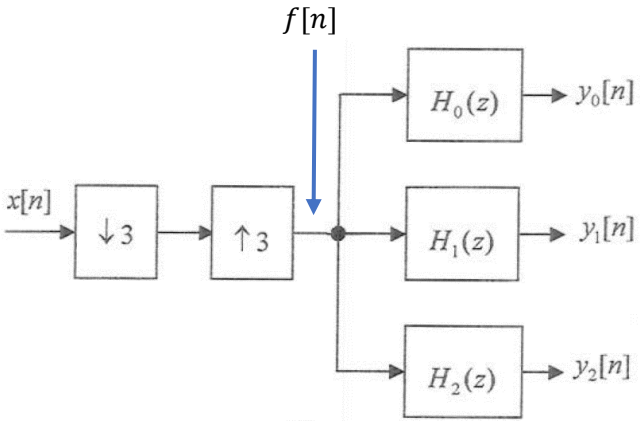
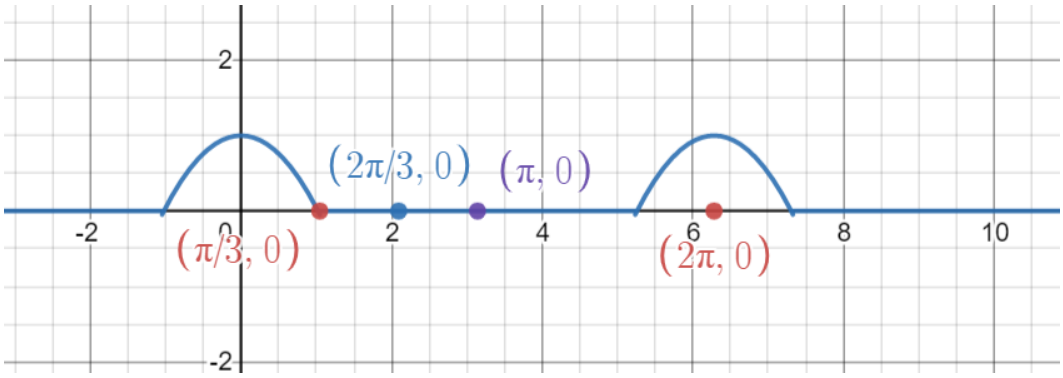


Figure 1

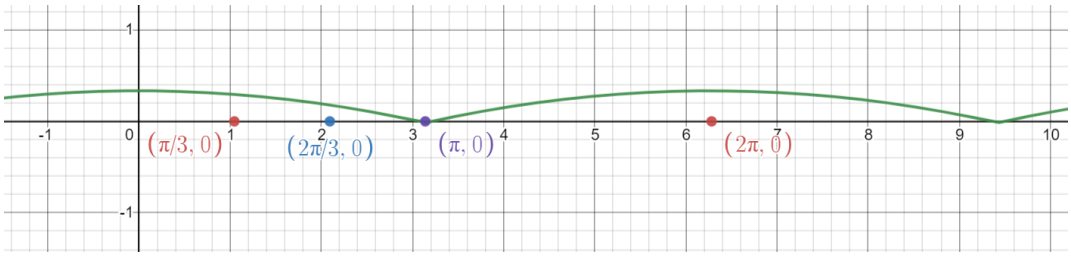
[15]

Answer

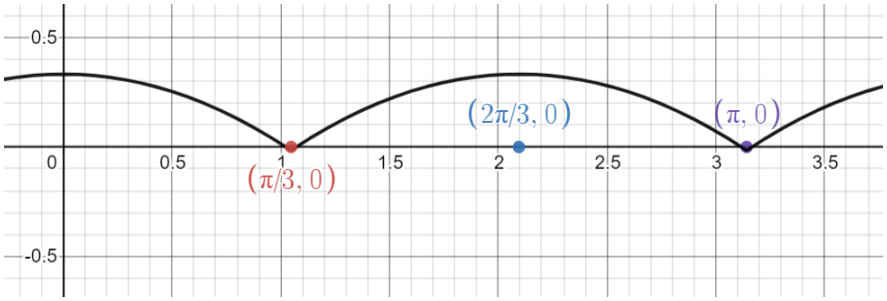
The DTFT of the input is shown below:



After decimation by 3:



After interpolation by 3 we obtain the DTFT of the signal $f[n]$ shown below.



The output of the three filters is obvious.

(b) Develop an expression for the output $y[n]$ as a function of the input $x[n]$ for the multirate structure of **Figure 2** below.

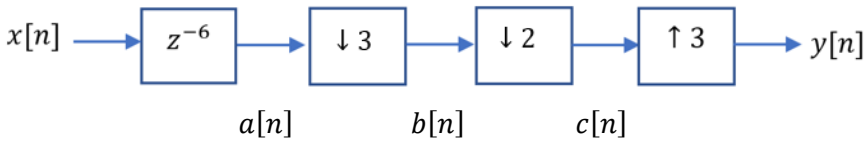


Figure 2

[10]

Answer

$a[n] = x[n-6]$
$b[n] = a[3n]$
$c[n] = b[2n] = a[6n] = x[6n-6]$
$y[n] = c\left[\frac{n}{3}\right] = x[2n-6]$